higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO INFORMATION SYSTEMS
NQF LEVEL 2

(10041002)

1 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 19 pages.

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TIME: 3 HOURS
MARKS: 200

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCEM

SECTION A

QUESTION 1

QUESTION 1.1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 The first step that a manager should take in problem solving is to …

A look at solutions.
B define the problem.
C look for the cause.
D evaluate the solution.

Questions 1.1.2 to 1.1.6 are based on the following figure (Figure 1), and description.

Assume the computer file named ***WeekTempratures.txt*** saved in the folder named ***WeatherApp*** on the **C: Drive** of a desktop computer contains the name of the week and the lowest and highest temperatures for the day.

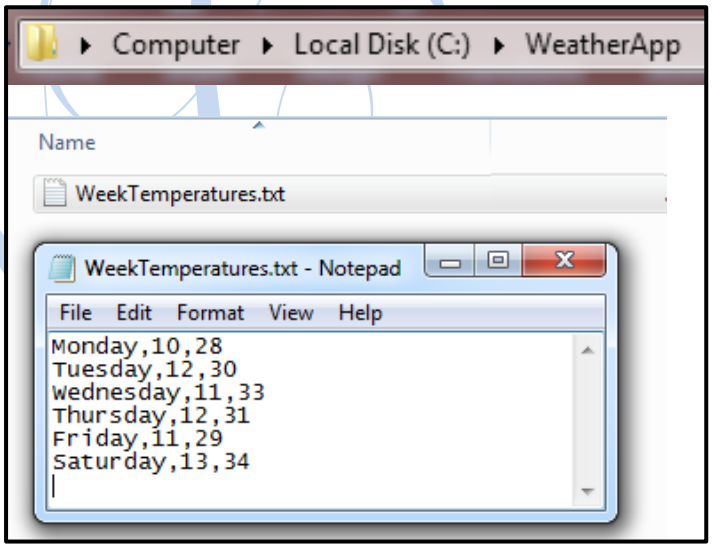


Figure 1

A certain computer application uses the text file to determine, the highest, lowest, and average temperatures for the week as well as the day that had the biggest change in temperature.

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1.1.2 Considering the concept of information processing, the purpose of the text file in this scenario is for …

A input
B output
C processing
D storage

1.1.3 The raw facts stored in the file is referred to as …

A intelligence.
B information.
C knowledge.
D data.

1.1.4 When the file is loaded for processing and the content is read from the file it would first be stored in the …

A RAM.
B ROM.
C CPU.
D ALU.

1.1.5 The **.txt** as part of the name of the file is called the file …

A code.
B abbreviation.
C signature.
D extension.

1.1.6 The Notepad application is an example of a / an …

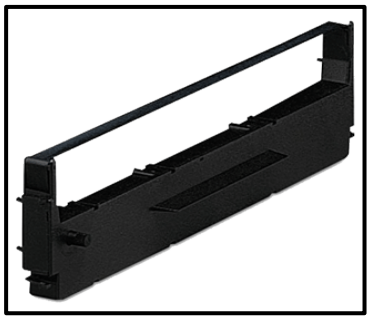
A Operating system
B Browser
C Utility software
D System software

1.1.7 To show the help menu in windows the ... key is used.

A F1
B F2
C F3
D F4

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1.1.8 Study the following image of a ribbon cartridge.



Which type of printer uses this type of cartridge?

A Laser
B Plotter
C Inkjet
D Dot Matrix

Questions 1.1.9 to 1.1.10 is based on the following figure (Figure 2) which represent a certain binary number.

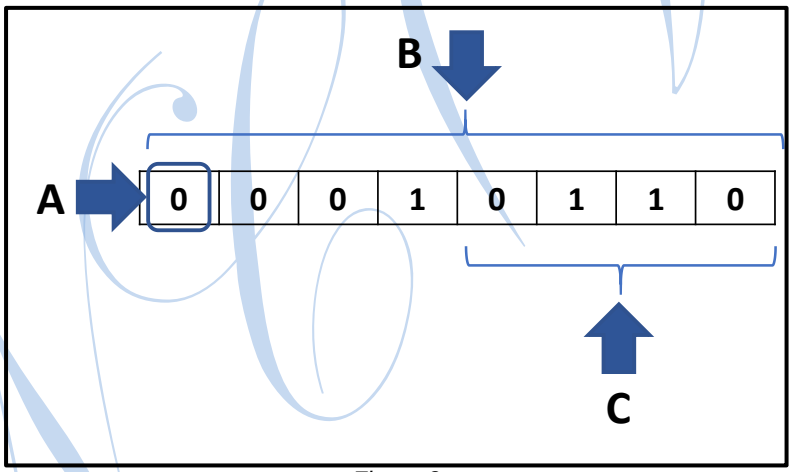


Figure 2

1.1.9 The value pointed to by arrow A is called a …

A Byte
B Bit
C Word
D Nibble

1.1.10 The 4 values represented by the arrow C has an equivalent decimal value of …

A 110
B 0110
C 6
D 4

(10 × 1) (10)

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QUESTION 1.2

1.2 Complete each of the following sentences by using a word from the table below. Write only the word next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

storage; knowledge; organization; system; application; master terminal; optical; mandatory; output; virus; pilot; memory

1.2.1 A/An … is considered to be a group of people who work together towards a common goal.

1.2.2 Information which is given context becomes …

1.2.3 … access control is a security strategy that restricts the ability individual resource owners have to grant or deny access to resource objects in a file system.

1.2.4 A/An … is a program that often interferes with computer systems or damages data.

1.2.5 Peripherals can be divided into three categories: input, output and … peripheral devices.

1.2.6 … software is also called end-user programs.

1.2.7 A computer … is an electromechanical hardware device which is used for entering data into and displaying data from a computer system.

1.2.8 … hardware such as a monitor is used to display the results of computer data that has been processed.

1.2.9 A/An … disc such as a DVD, stores digital data.

1.2.10 Files that typically contain descriptive data is referred to as … files.

(10 × 1) (10)

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QUESTION 1.3

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

1.3.1 Performing disk defragmentation is an important maintenance task for hard drives.

1.3.2 Information systems are used to help managers in decision making.

1.3.3 A computer virus can be spread through an infected e-mail attachment.

1.3.4 A shareware antivirus program allows the user to redistribute and sell the product.

1.3.5 A supercomputer is a powerful computer that can process large amounts of data very quickly

1.3.6 An automated teller machine is an example of an intelligent terminal.

1.3.7 A flash drive is an example of an input device.

1.3.8 The resolution of a computer screen is made up of tiny dots called electrons.

1.3.9 A typical HDD drive uses optical technology.

1.3.10 1 kilobyte is equal to 1 055 bytes.

(10 × 1) (10)

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QUESTION 1.4

1.4 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.4.1	Is a system used to define a hierarchy within an organisation.	A	procedures
1.4.2	The element of an information system which is used to guide employees in their roles	B	online advertisements
1.4.3	The action of identifying and removing potentially malicious software from a computer system.	C	plotter
1.4.4	A method used by companies to get customers to see their products on the internet	D	expansion boards
1.4.5	A type of processing where the information is captured but the actual processing is done at a later stage	E	compact disc case
1.4.6	Cards which are added to the motherboard to add additional functionality to a computer	F	input hardware
1.4.7	Hardware that is used to enter data into a computer	G	organisational structure
1.4.8	A low-speed printing device which is large and takes up a lot of space	H	data projector
1.4.9	A display device that uses a bright light to project a video	I	virus scan
1.4.10	A plastic container used to keep optical storage disks safe to avoid damage	J	batch processing
		K	backup
		L	Laser printer
		M	real-time processing

(10 × 1) (10)

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QUESTION 1.5

1.5 Write out the following acronyms in full next to the question number (1.5.1–1.5.10) in the ANSWER BOOK.

1.5.1 CEO

1.5.2 PC

1.5.3 ALU

1.5.4 DSS

1.5.5 AMD

1.5.6 DRAM

1.5.7 WMA

1.5.8 CAD

1.5.9 EFT

1.5.10 DVD

(10 × 1) (10)
[50]

TOTAL SECTION A: 50

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SECTION B

QUESTION 2

2.1 Read the following scenario and then answer the questions that follow.

John Mpanza, the CEO of Themmax Enterprises, a computer hardware and solutions company, has a monthly meeting with all the managers of the various departments.

The following departmental managers were instructed to attend the meeting for a presentation:

- ***Mpanza***, must prepare a report showing the monthly salary budget as well as the number of employees taking leave. As part of his presentation he also shows the advert for two new posts in the production department.
- ***Siphiwa*** must prepare a presentation on the proposed use of new biometric security technologies as part of some of the Laptops they manufacture,
- ***Sunitha*** is to show their latest YouTube and DSTV advertisements which promotes the launch of their new range of Laptops
- ***Hleza*** must prepare a report on production statistics for the month.
- ***Dlamane*** must prepare an annual report on the organisation's income and expenses for the year and well as present a forecast for the next twelve months sales.

Identify the departmental manager who holds which position listed below and motivate your answer.

2.1.1 Operations manager

2.1.2 Financial manager

2.1.3 Research and development manager

2.1.4 Human Resources (HR) manager

2.1.5 Marketing manager

(5 × 2) (10)

2.2 Define the term management. (2)

2.3 Differentiate between the terms: *methodical* and *analytical*, with regards to analysing information. (2 × 2) (4)

[16]

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QUESTION 3

Study the following diagram as presented in Figure 3, representing the typical components of an Information System. ***Note:*** Some components have been labelled with the letters **A** to **E**

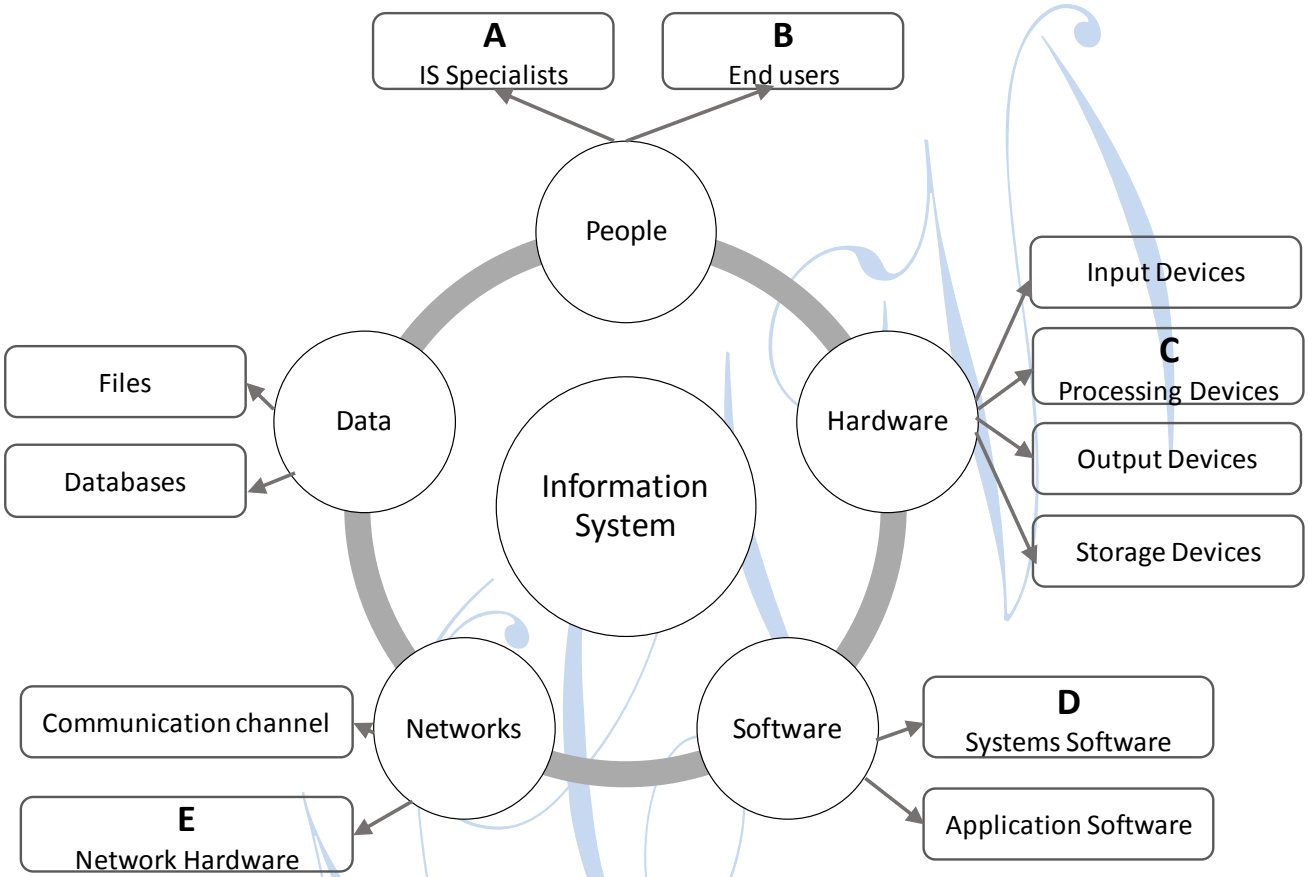


Figure 3

3.1 Write a short definition for each component labelled A to E. Write down the letter followed by your definition. (2 × 5) (10)

3.2 Write down the name of the components (labelled A to E) and supply an applicable, typical example of such a component.

E.g. Input Device – Keyboard
Application Software – Ms Word (1 × 5) (5)
[15]

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QUESTION 4

Study the following diagram as presented in Figure 4, representing a computerised information system with a single server to which all users connect to.

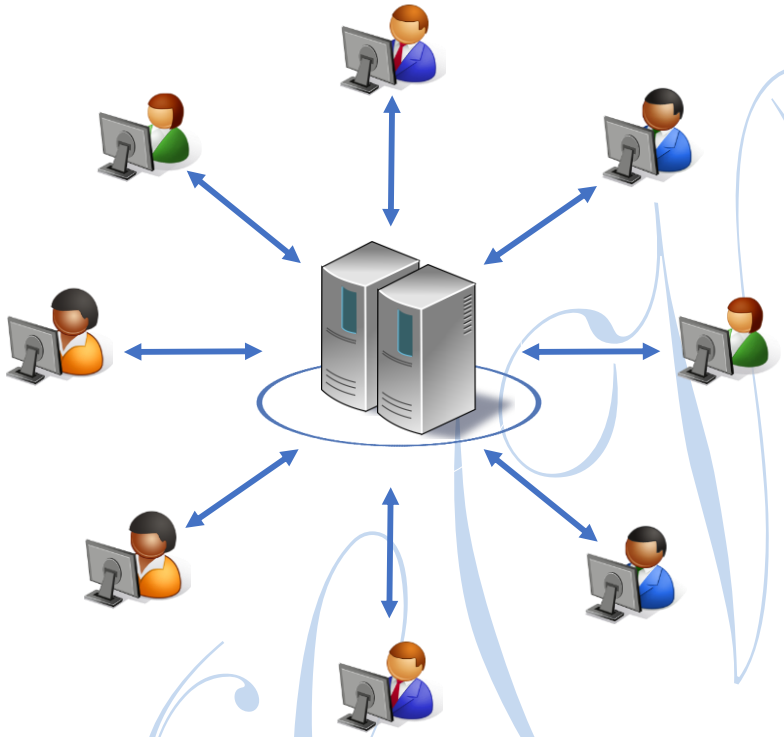


Figure 4

4.1 Identify the type of information system depicted in Figure 4. (1)

4.2 List three advantages of such a configuration (for the type of information system you have identified for question 4.1). (3)

4.3 List three disadvantages of such a configuration (for the type of information system you have identified for question 4.1). (3)

4.4 State the name of any antivirus software package. (1)

4.5 Differentiate between three types of viruses. State the name of the type of virus and a short description of each. (6)

4.6 Explain the need for keeping antivirus software updated. (2)

4.7 Using anti-virus software is one kind of control that the college could use to help secure their information system.

Name five other types of security related controls that the college may use to protect their information system. (5)

[21]

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QUESTION 5

5.1 Differentiate between general-purpose and special purpose computers.

(2 × 2) (4)

Study the following screenshot as presented in Figure 5 of a web application where customers are allowed to make online purchases. Figure 6 shows some additional specifications of the notebook. The current customer is busy viewing the following option as shown below.

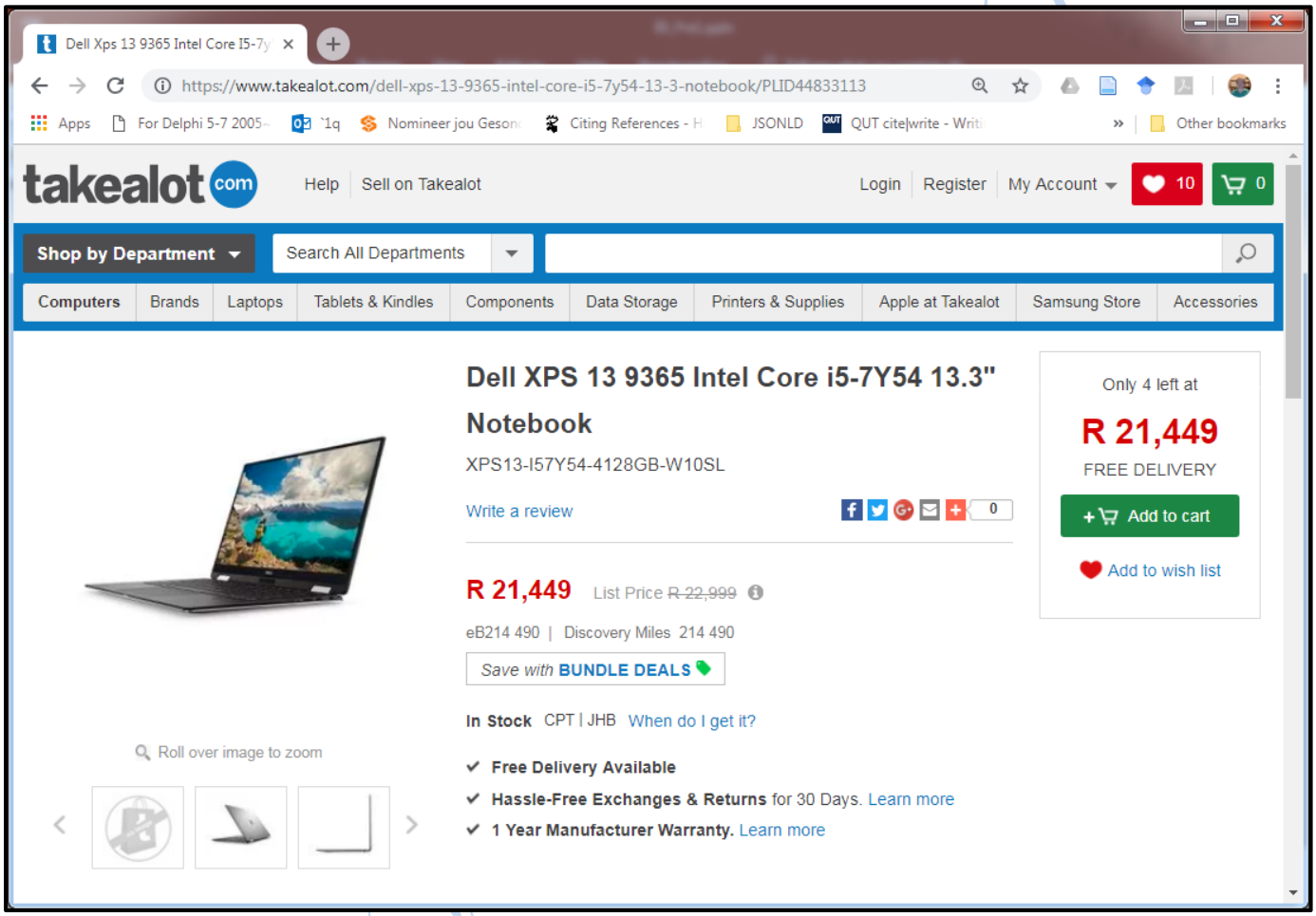


Figure 5

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Specifications:
Dell XPS 13 9365 Intel Core i5-7Y54 13.3" Notebook

Features:
- Intel Core i5-7Y54
- 13.3" FHD (1920 x 1080) touch
- 4GB LPDDR3 1866MHz, 128GB SSD
- HD Graphics 615
- Windows 10 Home (64bit) SL, 1YR NBD

Specifications:
- 7th Generation Intel Core i5-7Y54 Processor (4M cache, up to 3.2 GHz)
- 13.3" FHD (1920 x 1080) InfinityEdge touch display, Silver
- Finger Print Palmrest,82-key keyboard
- 30W Power Adapter

- Dell Adapter USB-C to USB-A 3.0
- Intel(R) Core(TM) i5 Label
- Intel(R) HD Graphics 615
- 2x2 WiFi and Bluetooth
- International English Backlit 82-key Keyboard
- Windows 10 Home (64bit) Single Language English
- Microsoft Office 30 Day Trial - Excludes Office License
- McAfee 30day Trial
- 1 Year Hardware Support - Next Business Day
- Warranty: 12 Months

Ports and other specifications of the **Dell XPS 13 9365 Intel Core i5-7Y54 13.3" Notebook**

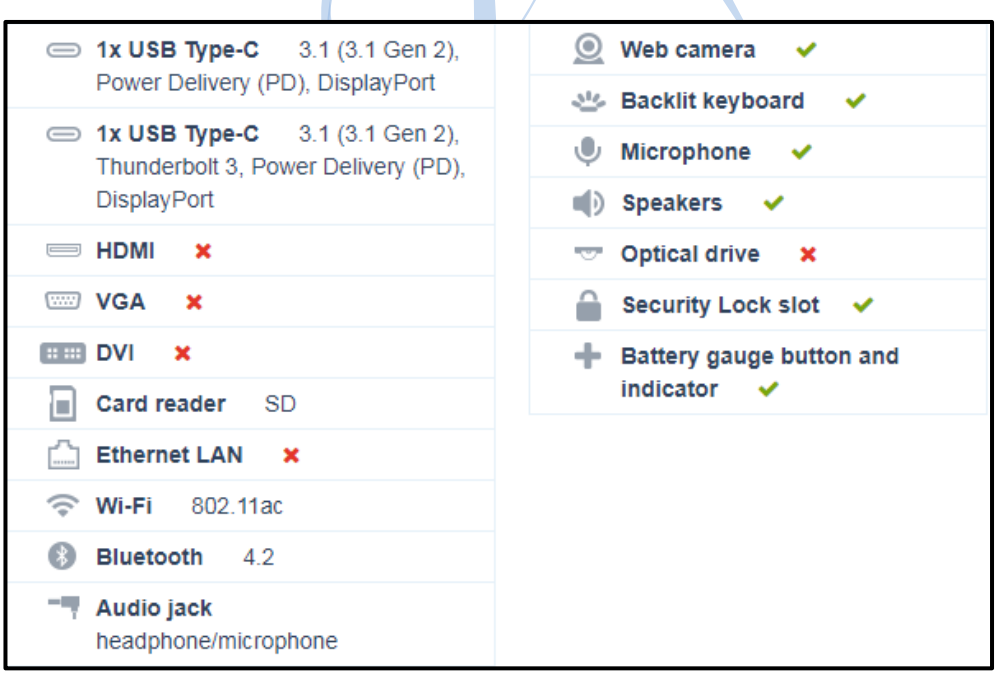


Figure 6

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5.2 5.2.1 State the name of the online retailer. (1)

5.2.2 Is the Notebook an example of a general purpose or a special purpose computer? (1)

5.2.3 Differentiate between a laptop and notebook computer. (2)

5.2.4 Identify the capacity of the secondary storage device. (1)

5.2.5 Expand the acronym: SSD. (1)

5.2.6 State the size of the screen. (1)

5.2.7 Would the user be able to listen to CDs by using the netbook? Motivate your answer. (1 + 1) (2)

5.2.8 How would the user be able to connect to a network? (1)

5.2.9 Except for the keyboard, list TWO other input devices that are part of the specifications of the notebook. (1 × 2) (2)

5.2.10 Except for the screen, identify ONE other output device that are part of the specifications of the notebook. (1)

5.2.11 Describe what the following specification entail:
Microsoft Office 30 Day Trial - Excludes Office License (2)

5.2.12 State the name and code of the component representing the CPU of the notebook. (2)

5.3 List the FOUR main steps of the CPU cycle. (4)
[25]

QUESTION 6

6.1 Study the images shown below.

DEVICE A	DEVICE B	DEVICE C
		

Name the devices A to C. Write only the answer next to the letter (A–C) in your ANSWER BOOK. (3 × 1) (3)

6.2 Differentiate between *keyboard entry* and *direct entry*. (2 × 2) (4)

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6.3 State the name given to the standard keyboard layout used in South Africa. (1)

6.4 Batch processing as depicted in the Figure 7 below is an example of one of the modes of processing that could be used as part of an information system.

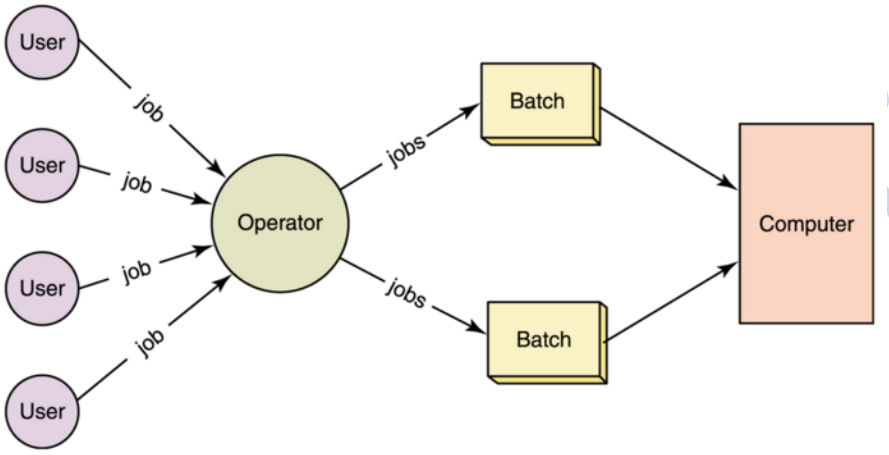


Figure 7

Batch processing: Source http://www.itrelease.com

Differentiate between two other types of processing and include an applicable example as well. (2 × 3) + (2 × 1) (8)

6.5 List TWO advantages an AGP card have over a PCI graphics card. (2)
[18]

QUESTION 7

7.1 Study the following image below and answer the questions that follow:

Samsung HM250HI 250GB SATA (M7) 2.5 inch



Figure 8

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7.1.1 Identify the hardware device as presented in Figure 8. (1)

7.1.2 State the type of media that is used to store data on the device. (1)

7.1.3 Explain how data is stored on this device. (4)

7.1.4 List any TRHEE disadvantages associated with the device as presented in Figure 8. (3)

7.1.5 State the name of the large circular component that is part of the device. (1)

7.2 List FIVE reasons why it is important for an organization to regularly perform backups. (5 × 1) (5)

7.3 List FOUR typical functions of a multi-function printer. (4 × 1) (4)

7.4 Bongi has a small business that helps people to create and edit a professional looking curriculum vitae.

Explain why a Laser printer would be a suitable choice for Bongi's business. (4)

7.5 What is a *Petabyte*? (2)

[25]

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QUESTION 8

8.1 Study the following image and description below and answer the questions that follow.



Figure 9 Android smart watch

DZ09 Android Bluetooth Smart Watch

Phone + Camera + SIM Card Slot Black + Silver
Single SIM Smart watch phone is an amazing product made to fit the new technology in the modern world. Made with copper colour the watch is so attractive to the eyes and the design used is spectaculars. It can fit the wrist of anybody as its arms are flexible and can be made to fit every hand. Everyone should own this watch as the new way to receive calls in the modern world.

Features:
533MHz Processor
GSM 850/900/1800/1900 MHz, single sim card
RAM: 128M+64M;32 GB Max. TF Storage Card
Bluetooth dialler, call reminder, Bluetooth phone
Built in Camera
Bluetooth SMS, instant messaging push
1.56 "TFT LCD touch screen 240 * 240 pixels
Music player
Calendar (can be synchronized)

8.1.1 The Android smart watch as presented in Figure 9 could be regarded as a computer.

Is this statement true or false? Provide a detailed motivation of your answer. (5)

8.1.2 Expand the acronym: LCD (1)

8.1.3 Briefly explain how LCD technology works (3)

8.1.4 Name another type of technology also used in flat panel displays such as that of the smart watch. (1)

8.1.5 Name ONE example of application software that is available with the watch (1)

8.1.6 TF Storage cards and SD storage cards are virtually similar.
Why would a user add such a card to the smart watch? (2)

8.1.7 Define the term: *Soft copy*. (2)

8.1.8 Categorise the primary output device of the smartwatch as either a hardcopy or softcopy device (1)

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8.2 Choose an item from COLUMN B **and** COLUMN C that matches a description in COLUMN A. Write only the letter (A–E) and the corresponding roman number (I-V) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	COLUMN C	
File description		**File Icon**	**File type**	
8.2.1	A computer file that contains an encoded sequence of instructions that the system can run directly when the user clicks the file icon.	A 	I	Power Point File format
8.2.2	A file format that has captured all the elements of a printed document as an electronic image that you can view, navigate, print, or forward to someone else.	B 	II	Zipped archive file
8.2.3	A file that stores electonic slides made by a user to use as part of a presentation.	C 	III	Executable
8.2.4	An archive file format that supports lossless data compression	D 	IV	Montion Picture Experts group file
8.2.5	A digital multimedia container file most commonly used to store video and audio	E 	V	Portable document format

(5 × 2) (10)

8.3 Differentiate between *impact* and *non-impact* printers (1 × 2) (2)

8.4 A laser printer is an example of a non-impact printer. List two other examples of non-impact printers. (2 × 1) (2)

[30]

TOTAL SECTION B: 150
GRAND TOTAL: 200

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